

Riley Brown

rydb@proton.me | rbrown71@nebraska.edu | ry3b.github.io | github.com/ry3b

EDUCATION

University of Nebraska-Lincoln

B.S. Mathematics and Data Science, Minor in Computer Science, Lincoln, NE | Aug 2023 - May 2027 (expected)

- GPA 3.92/4.00
- University Honors Program

Budapest Semesters in Mathematics

Study abroad, mathematics, Budapest, Hungary | Jan 2026 - Jun 2026

PREPRINTS

Fixed-Parameter Tractability of t -Uniform Hypergraphicality

Riley Brown, Istvan Miklos

arXiv preprint, 2026

arXiv

RESEARCH EXPERIENCE

Undergraduate Researcher - Quantum LDPC Codes

University of Nebraska-Lincoln, Lincoln, NE | Fall 2026

Research Assistant, pseudorandomness

North Carolina State University, Raleigh, NC | Summer 2026

- Worked through Ta-Shma's epsilon-balanced code construction and the surrounding literature as a research partner with Prof. Chin Ho Lee
- Found simpler constructions of epsilon-balanced codes and minor improvements to the Hsieh-Mohanty-Zheng construction

Research Assistant, 3-uniform hypergraphic degree sequences, FPT algorithm feasibility

Budapest Semesters in Mathematics, Budapest, Hungary | Spring 2026

- Proved t -uniform hypergraphicality is fixed-parameter tractable in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via a bounded-dimension spectrum representation and balancing hinge-flips
- Co-authored the resulting preprint, arXiv:2606.08523

Summer Researcher, algorithm design

Harvey Mudd College, Claremont, CA | Summer 2025

- Analyzed and improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics, with a team of four
- Improved branching strategies and base-case parameters for graph-theoretic reductions

Undergraduate Researcher, tiling theory

University of Nebraska-Lincoln, Lincoln, NE | Summer 2024

SKILLS

AI for mathematics: Lean 4 and Mathlib verification tooling, LLM-assisted proof search, agentic research harnesses

Languages: Python, MATLAB, Java, SQL, LaTeX

RELEVANT COURSEWORK

Mathematics: Bijective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Graph Theory, Algebra I, Coding Theory

Computer Science: Development and Analysis of Algorithms, Automata, Computation, and Language, Data Structures and Algorithms